2025 IEPR California Energy Demand Forecast LSE/BA Tables and RASS Kick Off

February 6, 2026

2027 Residential Appliance Saturation Study

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February 6, 2026

Background: What is the "RASS"?

RASS = **R**esidential **A**ppliance **S**aturation **S**tudy

- Comprehensive look at residential energy use
- Collects information from residents via survey
  - appliances
  - heating and cooling equipment
  - electric vehicles
  - general usage patterns

California Code of Regulations

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Cal. Code Regs. Tit. 20, § 1343 - Energy End User Data: Survey Plans, Surveys, and Reports

State Regulations Compare

(a) Each UDC that has experienced a peak electricity demand of 1000 MW or more in both the two calendar years preceding the required data filing date, and each natural gas utility that has delivered 100 billion cubic feet of gas per year in both of the two calendar years preceding the required data filing date shall complete the survey plans, surveys, and reports described in this Section, unless exempt as described under the Compliance Option described under subsection (f).

(b) Survey Plans and Plan Approval.

(1) Submittal of Survey Plans. For each survey a utility or UDC is required to perform under this Section, the utility or UDC must complete and submit to the Commission a plan for conducting the survey that is consistent with

Source: **California Code of Regulations Title 20, § 1343**

RASS Background

- Supports CEC data products
  - residential sector model
  - energy demand forecast

- Conducted periodically - three previous studies (2019, 2009, 2003)

- **Unit energy consumption** (UEC) - the amount of energy a single appliance is estimated to use in one year

- **Saturations** - percentage of households owning a specific appliance

RASS History

- Conducted by hired contractor in bidding process
- DNV GL/KEMA contracted for RASS 2019, 2009 and 2003

2019 RASS Participating Utilities:

- PG&E
- SDG&E
- SCE
- SoCalGas
- LADWP
- SMUD

High Priority RASS Topics

- ✓ Electrification - heat pumps
- ✓ Residential solar
- ✓ Residential storage
- ✓ Electric vehicles
- ✓ Equity
- ✓ Work from Home

Timeline

- Begin utility engagement – Q1 2026

- Draft Request for Proposal (RFP) – Q1 & Q2 2026
- Release RFP – Q2 2026
- Award RFP – Q3 2026

- Survey implementation – Q3 2027

- Finalize study – 2029

Next Steps

- RFP
- Utility engagement
  - Follow up meetings
  - RASS data needs

Thank You!

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Open Discussion

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Load Serving Entity/Balancing Authority Tables

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Demand Analysis Branch, Energy Assessments Division

February 6, 2026

2025 IEPR Forecast Framework

	Planning Forecast	Local Reliability w/o Known Loads	Local Reliability w/ Known Loads
Economic, Demographic, and Price Scenarios	Baseline	Baseline	Baseline
Data Centers	Mid	High	High
Known Loads	Excluded	Excluded	Included
BTM PV and Storage	Mid	Low	Low
Additional Achievable Energy Efficiency (AAEE) Scenario	Scenario 3	Scenario 2	Scenario 2
Additional Achievable Fuel Substitution (AAFS) Scenario	Scenario 2	Scenario 3	Scenario 3
Additional Achievable Transportation Electrification (AATE) Scenario	Scenario 2	Scenario 3	Scenario 3

Source: CEC Staff

Example Use Cases

- Single Forecast Set Agreement between CEC, CAISO, CPUC documents agreement on use of various forecast elements

Planning Forecast	Local Reliability w/o Known Loads	Local Reliability w/ Known Loads	2024 IEPR Planning
Resource Adequacy	CAISO Local Capacity Technical Studies	CAISO TPP Local Area Reliability Studies	CPUC IRP Reference and Preferred System Plans
CAISO Maximum Import Capability Studies	-	-	CAISO TPP Economic Studies
CAISO Flexible Capacity Studies	-	-	-

Source: CEC Staff

- See the Single Forecast Set Agreement for additional detail:
    - 25-IEPR-03, TN # 268288, January 21, 2026

Load Serving Entity and Balancing Authority Forms

- These forms are disaggregation of the adopted forecast
- Forms themselves are not adopted
- Forms included:
  - Form 1.1c: Electricity Deliveries to End Users by Agency (GWh)
  - Form 1.5a: Total Energy to Serve Load by Agency and Balancing Authority (GWh)
  - Form 1.5b: 1-in-2 Net Electricity Peak Demand by Agency and Balancing Authority (MW)
  - Form 1.5c: 1-in-5 Net Electricity Peak Demand by Agency and Balancing Authority (MW)
  - Form 1.5d: 1-in-10 Net Electricity Peak Demand by Agency and Balancing Authority (MW)
  - Form 1.5e: 1-in-20 Net Electricity Peak Demand by Agency and Balancing Authority (MW)
- Draft Forms are posted for comment on the DAWG event page

Form 1.1c Annual End User Deliveries by Agency

Annual End User Deliveries by Agency

- First, Utility Distribution Company forecasts are constructed starting with the baseline and managed sales forecasts
  - Remove loads with different growth rates from forecasts
    - Pumping loads, data centers & known loads
  - Calculate year-over-year growth rates for each sector and forecast zone
  - Apply growth rates to LSE sales from last historical year
    - i.e., each LSE within a forecast zone has the same growth rate
  - Add pumping load, data centers, and known loads back to the specific responsible LSEs
  - Check that aggregated LSE forecasts match the original zone-level forecasts

CCA Sales Forecast Methodology

- IOU UDC forecasts are disaggregated to IOU, individual CCAs, and aggregate ESP
- Start with CCA historic and near-term forecasts as adjusted for CCA expansion plans
  - 2023 QFER 1306b data
  - CCA Implementation plan filings
  - 2025 IRP data submittals
  - CED 2025 demand form filings
- Remove loads with different growth rates from IOU UDC forecasts
  - Data centers & known loads
- Calculate year-over-year growth rates for each sector and forecast zone
- Apply growth rates to CCA sales from last historical or near-term forecast year
  - i.e., each CCA sector within a forecast zone has the same growth rate
- Add data centers and known loads back to the specific responsible CCAs

Allocation of Data Center Load to LSEs

- Lists of data center projects are received from relevant utilities to create the forecast
- December data request asked for utilities to include "LSE responsible for procurement"
- Utilities assigned LSEs/CCAs based on location
- PG&E also identified some projects as ESP or potentially ESP-served
- Opt-out rates were applied to CCA identified projects:
  - PGE – 13% per 2026 ERRA documentation
    - "Potentially ESP" projects were allocated 50/50 between the candidate CCA and ESPs
  - SCE -30% based on preliminary analysis by SCE
    - 5% allocated to direct access
  - SDGE - CCAs not yet identified, so load allocated based on commercial sector share

Allocation of Known Loads to LSEs

- LSE information requested from utilities as part of Known Loads data request
- However, many records in the Known Loads dataset did not identify an LSE
  - PG&E
    - ~36% of records missing LSE
  - SCE
    - ~61% of records missing LSE
  - SDG&E
    - ~86% of records missing LSE
  - LSEs assigned by CEC staff by examining list of counties/cities served by a CCA
- Opt-out rates consistent with assumptions used for data center forecast, but no loads were allocated to direct access.

Form 1.5a

Total Energy to Serve Load by Agency and Balancing Authority

- This form adds energy losses and allocates load to Local Capacity Areas (LCAs)
  - TAC area totals are determined by the Hourly Load Model annual energy forecast
  - Agency Retail Deliveries are mapped to zones and loss factors are applied:

Planning Area	Energy Loss Factor	Pumping Load Loss Factor
PGE	1,091	1,086
SCE	1,068	1,086
NCNC	1,064	1
LADWP	1,135	1
BUGL	1,064	1
IID	1,128	1

Source: CEC Staff

- CAISO EMS “subpoena’ data are used in combination with the CEC zone forecast to disaggregate PGE loads to Bay Area and ZP 26
- SCE hourly substation loads are used to calibrate allocation of CEC zone forecasts to SCE LCAs (Big Creek/Ventura and Los Angeles Basin)
- Data centers and known loads were allocated to LCAs based on IOU-provided data

Form 1.5b

1-in-2 Net Electricity Peak Demand by Agency and Balancing Authority

- Each TAC or BA total shown is the TAC noncoincident peak
  - The TAC area totals are determined by the Hourly Load Model annual peaks
  - 2025 TAC peaks are weather-normalized summer 2025 peaks
  - For LSEs within CAISO, an LSE-specific peak-to-energy ratios and the TAC weather adjustment factor is applied to Form 1.5a energy.
- For non-CAISO BA's, 2025 values are actual 2025 peak.
- Peak-to-energy ratios calculated from CAISO and SCE hourly loads are applied to the LCA energy forecasts, excluding data centers and known loads, to allocate TAC load to zones
- Data center and known load peaks are allocated to LCAs based on IOU-provided data.

Form 1.5c- Form 1.5e

1-in-5, -10 and -20 Net Electricity Peak Demand by Agency and Balancing Authority

- Extreme temperature scalars based on staff weather-normalization analysis are applied to the 1-in-2 forecast
- Scalars are not applied to pumping loads, so scalars on remaining load within the PGE and SCE TACs are adjusted up
- Scalars were not updated for this forecast

Planning Area	1-in-5	1-in-10	1-in-20
PGE	1.051	1.079	1.106
SCE	1.065	1.094	1.120
SDGE	1.047	1.076	1.105
NCNC	1.044	1.090	1.129
LADWP	1.060	1.096	1.122
IID	1.037	1.046	1.051
BUGL	1.079	1.112	1.125
SMUD	1.044	1.090	1.129
VEA	1.065	1.094	1.120

Source: CEC Staff

2024 IEPR Planning Forecast Use Case

- Only change will be to LSE forecasts within the IOU distribution service areas (Form 1.1c)
  - Starting point of IOU bundled and CCA forecasts will be updated with near-term data on CCA load migration and recorded data
  - 2025 LSE data center allocations will be applied to 2024 IEPR data center sales

- No change to the Form 1.5's

Comments

- Please send comments and questions to CEC staff by Tuesday, February 17, 2026

- General comments or questions on LSE/BA tables:
  - Contact Lynn Marshall
    Lynn.Marshall@energy.ca.gov

- Comments or questions on data centers:
  - Contact Mathew Cooper
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- Comments or questions on known loads:
  - Contact Asish Gautam
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Open Discussion

Heidi Javanbakht

Thank you!

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